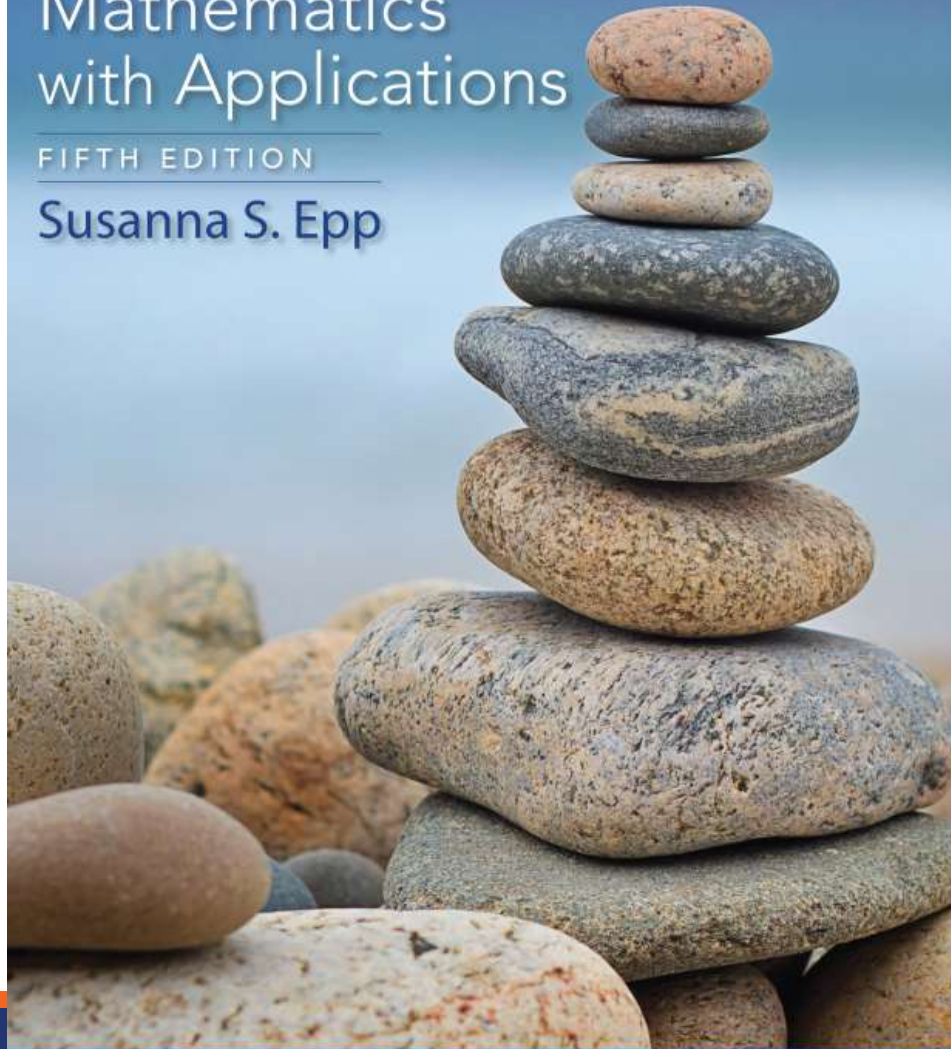


Discrete Mathematics with Applications

FIFTH EDITION

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Faculty of Computer Science

UFCFFS-30-1– Foundations of Computing: Mathematic Induction & Recursion

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Learning Objectives

1. Describe the features of a queue and the operations on it
2. Choose a queue implementation based on its performance characteristics
3. Recognize applications where it is appropriate to use a queue
4. Explain the difference between a queue and a priority queue
5. Recognize applications where it is appropriate to use a priority queue

Reading

- ❑ Chapter 5 - Discrete Mathematics with Applications 5th Edition (Susanna's Book)
- ❑ Chapter 5 – Discrete Mathematics and Its Applications 7th Edition (Kenneth Rosen's book)

Mathematical Induction I: proving Formulas

Principle of Mathematical Induction

Let $P(n)$ be a property that is defined for integers n , and let a be a fixed integer. Suppose the following two statements are true:

1. $P(a)$ is true.
2. For every integer $k \geq a$, if $P(k)$ is true then $P(k + 1)$ is true.

Then the statement

for every integer $n \geq a$, $P(n)$

is true.

Method of Proof by Mathematical Induction

Consider a statement of the form, “For every integer $n \geq a$, a property $P(n)$ is true.”
To prove such a statement, perform the following two steps:

Step 1 (basis step): Show that $P(a)$ is true.

Step 2 (inductive step): Show that for every integer $k \geq a$, if $P(k)$ is true then $P(k + 1)$ is true. To perform this step,

suppose that $P(k)$ is true, where k is any
particular but arbitrarily chosen integer with $k \geq a$.

*[This supposition is called the **inductive hypothesis**.]*

Then

show that $P(k + 1)$ is true.

Sum of the First n Integers

Use mathematical induction to prove that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for every integer } n \geq 1.$$

Solution To construct a proof by induction, you must first identify the property $P(n)$. In this case, $P(n)$ is the equation

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}. \quad \leftarrow \text{the property } (P(n))$$

[To see that $P(n)$ is a sentence, note that its subject is “the sum of the integers from 1 to n ” and its verb is “equals.”]

In the basis step of the proof, you must show that the property is true for $n = 1$, or, in other words, that $P(1)$ is true. Now $P(1)$ is obtained by substituting 1 in place of n in $P(n)$. The left-hand side of $P(1)$ is the sum of all the successive integers starting at 1 and ending at 1. This is just 1. Thus $P(1)$ is

$$1 = \frac{1(1+1)}{2}.$$

← basis ($P(1)$)

Of course, this equation is true because the right-hand side is

$$\frac{1(1+1)}{2} = \frac{1 \cdot 2}{2} = 1,$$

which equals the left-hand side.

In the inductive step, you assume that $P(k)$ is true, for a particular but arbitrarily chosen integer k with $k \geq 1$. *[This assumption is the inductive hypothesis.]* You must then show that $P(k+1)$ is true. What are $P(k)$ and $P(k+1)$? $P(k)$ is obtained by substituting k for every n in $P(n)$. Thus $P(k)$ is

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

← inductive hypothesis ($P(k)$)

Similarly, $P(k+1)$ is obtained by substituting the quantity $(k+1)$ for every n that appears in $P(n)$. Thus $P(k+1)$ is

$$1 + 2 + \cdots + (k+1) = \frac{(k+1)((k+1)+1)}{2},$$

or, equivalently,

$$1 + 2 + \cdots + (k+1) = \frac{(k+1)(k+2)}{2}. \quad \leftarrow \text{to show } (P(k+1))$$

Now the inductive hypothesis is the supposition that $P(k)$ is true. How can this supposition be used to show that $P(k+1)$ is true? $P(k+1)$ is an equation, and the truth of an

left-hand and right-hand sides until you see that they are the same. In this case, the left-hand side of $P(k+1)$ is

$$1 + 2 + \cdots + (k+1),$$

which equals

$$(1 + 2 + \cdots + k) + (k+1)$$

By substitution from the inductive hypothesis,

$$\begin{aligned} & (1 + 2 + \cdots + k) + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \\ &= \frac{k^2 + k}{2} + \frac{2k + 2}{2} \\ &= \frac{k^2 + 3k + 2}{2} \end{aligned}$$

The next-to-last term is k because the terms are successive integers and the last term is $k+1$.

since the inductive hypothesis says that
 $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$

by multiplying the numerator and denominator
of the second term by 2 to obtain a common
denominator

by multiplying out the two numerators

by adding fractions with the same denominator
and combining like terms.

So the left-hand side of $P(k+1)$ is $\frac{k^2+3k+2}{2}$. Now the right-hand side of $P(k+1)$ is

$$\frac{(k+1)(k+2)}{2} = \frac{k^2+3k+2}{2} \quad \text{by multiplying out the numerator.}$$

Thus the two sides of $P(k+1)$ are equal to each other, and so the equation $P(k+1)$ is true.

Theorem 5.2.1 Sum of the First n Integers

For every integer $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof (by mathematical induction): Let the property $P(n)$ be the equation

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}. \quad \leftarrow P(n)$$

Show that $P(1)$ is true:

To establish $P(1)$, we must show that

$$1 = \frac{1(1+1)}{2}. \quad \leftarrow P(1)$$

But the left-hand side of this equation is 1 and the right-hand side is

$$\frac{1(1+1)}{2} = \frac{2}{2} = 1$$

also. Hence $P(1)$ is true.

Show that for every integer $k \geq 1$, if $P(k)$ is true then $P(k+1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 1$. That is:]

Suppose that k is any integer with $k \geq 1$ such that

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}. \quad \begin{array}{l} \leftarrow P(k) \\ \text{inductive hypothesis} \end{array}$$

[We must show that $P(k+1)$ is true. That is:] We must show that

$$1 + 2 + 3 + \cdots + (k+1) = \frac{(k+1)[(k+1)+1]}{2},$$

or, equivalently, that

$$1 + 2 + 3 + \cdots + (k+1) = \frac{(k+1)(k+2)}{2}. \quad \leftarrow P(k+1)$$

[We will show that the left-hand side and the right-hand side of $P(k+1)$ are equal to the same quantity and thus are equal to each other.]

The left-hand side of $P(k+1)$ is

$$\begin{aligned}
 &1 + 2 + 3 + \cdots + (k+1) \\
 &= 1 + 2 + 3 + \cdots + k + (k+1) && \text{by making the next-to-last} \\
 &= \frac{k(k+1)}{2} + (k+1) && \text{term explicit} \\
 &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} && \text{by substitution from the} \\
 &= \frac{k^2 + k}{2} + \frac{2k + 2}{2} && \text{inductive hypothesis} \\
 &= \frac{k^2 + 3k + 2}{2} && \text{by algebra.}
 \end{aligned}$$

And the right-hand side of $P(k + 1)$ is

$$\frac{(k + 1)(k + 2)}{2} = \frac{k^2 + 3k + 2}{2}.$$

Thus the two sides of $P(k + 1)$ are equal to the same quantity and so they are equal to each other. Therefore, the equation $P(k + 1)$ is true *[as was to be shown]*.

[Since we have proved both the basis step and the inductive step, we conclude that the theorem is true.]

Theorem 5.2.2 Sum of a Geometric Sequence

Theorem 5.2.2 Sum of a Geometric Sequence

For any real number r except 1, and any integer $n \geq 0$,

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}.$$

Proof (by mathematical induction): Suppose r is a particular but arbitrarily chosen real number that is not equal to 1, and let the property $P(n)$ be the equation

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}. \quad \leftarrow P(n)$$

We must show that $P(n)$ is true for every integer $n \geq 0$. We do this by mathematical induction on n .

Show that $P(0)$ is true:

To establish $P(0)$, we must show that

$$\sum_{i=0}^0 r^i = \frac{r^{0+1} - 1}{r - 1}. \quad \leftarrow P(0)$$

The left-hand side of this equation is $r^0 = 1$ and the right-hand side is

$$\frac{r^{0+1} - 1}{r - 1} = \frac{r - 1}{r - 1} = 1$$

also because $r^1 = r$ and, since $r \neq 1$, $r - 1 \neq 0$. Hence $P(0)$ is true.

Show that for every integer $k \geq 0$, if $P(k)$ is true then $P(k + 1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 0$. That is:]

Let k be any integer with $k \geq 0$, and suppose that

$$\sum_{i=0}^k r^i = \frac{r^{k+1} - 1}{r - 1} \quad \leftarrow P(k) \text{ inductive hypothesis}$$

[We must show that $P(k + 1)$ is true. That is:] We must show that

$$\sum_{i=0}^{k+1} r^i = \frac{r^{(k+1)+1} - 1}{r - 1},$$

or, equivalently, that

$$\sum_{i=0}^{k+1} r^i = \frac{r^{k+2} - 1}{r - 1}. \quad \leftarrow P(k + 1)$$

[We will show that the left-hand side of $P(k+1)$ equals the right-hand side.]

The left-hand side of $P(k+1)$ is

$$\begin{aligned}
 \sum_{i=0}^{k+1} r^i &= \sum_{i=0}^k r^i + r^{k+1} && \text{by writing the } (k+1)\text{st term separately from the first } k \text{ terms} \\
 &= \frac{r^{k+1} - 1}{r - 1} + r^{k+1} && \text{by substitution from the inductive hypothesis} \\
 &= \frac{r^{k+1} - 1}{r - 1} + \frac{r^{k+1}(r - 1)}{r - 1} && \text{by multiplying the numerator and denominator of the second term by } (r - 1) \text{ to obtain a common denominator} \\
 &= \frac{(r^{k+1} - 1) + r^{k+1}(r - 1)}{r - 1} && \text{by adding fractions} \\
 &= \frac{r^{k+1} - 1 + r^{k+2} - r^{k+1}}{r - 1} && \text{by multiplying out and using the fact that } r^{k+1} \cdot r = r^{k+1} \cdot r^1 = r^{k+2} \\
 &= \frac{r^{k+2} - 1}{r - 1} && \text{by canceling the } r^{k+1}\text{'s.}
 \end{aligned}$$

which is the right-hand side of $P(k+1)$ *[as was to be shown]*.

[Since we have proved the basis step and the inductive step, we conclude that the theorem is true.]

EXERCISE SET 5.2

1. Use the technique illustrated at the beginning of this section to show that the statements in (a) and (b) are true.

a. If $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right) = \frac{1}{5}$ then

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{6}\right) = \frac{1}{6}.$$

b. If $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{6}\right) = \frac{1}{6}$ then

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{6}\right)\left(1 - \frac{1}{7}\right) = \frac{1}{7}.$$

2. For each positive integer n , let $P(n)$ be the formula

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

- Write $P(1)$. Is $P(1)$ true?
- Write $P(k)$.
- Write $P(k + 1)$.
- In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, what must be shown in the inductive step?

3. For each positive integer n , let $P(n)$ be the formula

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n + 1)(2n + 1)}{6}.$$

- Write $P(1)$. Is $P(1)$ true?
- Write $P(k)$.
- Write $P(k + 1)$.
- In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, what must be shown in the inductive step?

4. For each integer n with $n \geq 2$, let $P(n)$ be the formula

$$\sum_{i=1}^{n-1} i(i + 1) = \frac{n(n - 1)(n + 1)}{3}.$$

- Write $P(2)$. Is $P(2)$ true?
- Write $P(k)$.
- Write $P(k + 1)$.
- In a proof by mathematical induction that the formula holds for every integer $n \geq 2$, what must be shown in the inductive step?

Mathematical Induction II: Applications

Proposition 5.3.2

For each integer $n \geq 0$, $2^{2n} - 1$ is divisible by 3.

Proof (by mathematical induction): Let the property $P(n)$ be the sentence “ $2^{2n} - 1$ is divisible by 3.”

$$2^{2n} - 1 \text{ is divisible by 3.} \quad \leftarrow P(n)$$

Show that $P(0)$ is true:

To establish $P(0)$, we must show that

$$2^{2 \cdot 0} - 1 \text{ is divisible by 3.} \quad \leftarrow P(0)$$

But

$$2^{2 \cdot 0} - 1 = 2^0 - 1 = 1 - 1 = 0,$$

and 0 is divisible by 3 because $0 = 3 \cdot 0$. Hence $P(0)$ is true.

Show that for any integer $k \geq 0$, if $P(k)$ is true then $P(k + 1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 0$. That is:]

Let k be any integer with $k \geq 0$, and suppose that

$$2^{2k} - 1 \text{ is divisible by } 3. \quad \leftarrow P(k) \text{ inductive hypothesis}$$

By definition of divisibility, this means that

$$2^{2k} - 1 = 3r \quad \text{for some integer } r.$$

[We must show that $P(k + 1)$ is true. That is:] We must show that

$$2^{2(k+1)} - 1 \text{ is divisible by } 3. \quad \leftarrow P(k + 1)$$

Now

$$\begin{aligned} 2^{2(k+1)} - 1 &= 2^{2k+2} - 1 \\ &= 2^{2k} \cdot 2^2 - 1 && \text{by the laws of exponents} \\ &= 2^{2k} \cdot 4 - 1 \\ &= 2^{2k}(3 + 1) - 1 \\ &= 2^{2k} \cdot 3 + (2^{2k} - 1) && \text{by the laws of algebra} \\ &= 2^{2k} \cdot 3 + 3r && \text{by inductive hypothesis} \\ &= 3(2^{2k} + r) && \text{by factoring out the 3.} \end{aligned}$$

But $2^{2k} + r$ is an integer because it is a sum of products of integers, and so, by definition of divisibility, $2^{2(k+1)} - 1$ is divisible by 3 *[as was to be shown]*.

[Since we have proved the basis step and the inductive step, we conclude that the proposition is true.]

Proposition 5.3.3

For every integer $n \geq 3$, $2n + 1 < 2^n$.

Proof (by mathematical induction):

Let the property $P(n)$ be the inequality

$$2n + 1 < 2^n. \quad \leftarrow P(n)$$

Show that $P(3)$ is true:

To establish $P(3)$, we must show that

$$2 \cdot 3 + 1 < 2^3. \quad \leftarrow P(3)$$

Now

$$2 \cdot 3 + 1 = 7 \quad \text{and} \quad 2^3 = 8 \quad \text{and} \quad 7 < 8.$$

Hence $P(3)$ is true.

Show that for every integer $k \geq 3$, if $P(k)$ is true then $P(k + 1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 3$. That is:]

Suppose that k is any integer with $k \geq 3$ such that

$$2k + 1 < 2^k. \quad \begin{array}{l} \leftarrow P(k) \\ \text{inductive hypothesis} \end{array}$$

[We must show that $P(k + 1)$ is true. That is:] We must show that

$$2(k + 1) + 1 < 2^{(k+1)}. \quad \leftarrow P(k+1)$$

Now

$$\begin{aligned} 2(k + 1) + 1 &= 2k + 1 + 2 && \text{by algebra} \\ &< 2^k + 2 && \text{because, by the inductive} \\ &&& \text{hypothesis, } 2k + 1 < 2^k \\ &< 2^k + 2^k && \text{because } 2 < 2^k \text{ since } k \geq 3 \\ &= 2 \cdot 2^k && \text{by algebra} \\ &= 2^{k+1} && \text{by the laws of exponents.} \end{aligned}$$

Thus by transitivity of order $2(k + 1) + 1 < 2^{k+1}$ [as was to be shown].

[Since we have proved the basis step and the inductive step, we conclude that the proposition is true.]

Example 5.3.3

Proving a Property of a Sequence

Define a sequence $a_1, a_2, a_3, \dots$ as follows:*

$$a_1 = 2$$

$$a_k = 5a_{k-1} \quad \text{for every integer } k \geq 2.$$

- Write the first four terms of the sequence.
- It is claimed that for each integer $n \geq 0$, the n th term of the sequence has the same value as that given by the formula $2 \cdot 5^{n-1}$. In other words, the claim is that the terms of the sequence satisfy the equation $a_n = 2 \cdot 5^{n-1}$. Prove that this is true.

Solution

a. $a_1 = 2$

$$a_2 = 5a_{2-1} = 5a_1 = 5 \cdot 2 = 10$$

$$a_3 = 5a_{3-1} = 5a_2 = 5 \cdot 10 = 50$$

$$a_4 = 5a_{4-1} = 5a_3 = 5 \cdot 50 = 250$$

- b. To use mathematical induction to show that every term of the sequence satisfies the equation, begin by showing that the first term of the sequence satisfies the equation. Then suppose that an arbitrarily chosen term a_k satisfies the equation and prove that the next term a_{k+1} also satisfies the equation.

Proof (by mathematical induction):

Let $a_1, a_2, a_3, \dots$ be the sequence defined by specifying that $a_1 = 2$ and $a_k = 5a_{k-1}$ for every integer $k \geq 2$, and let the property $P(n)$ be the equation

$$a_n = 2 \cdot 5^{n-1}. \quad \leftarrow P(n)$$

We will use mathematical induction to prove that for every integer $n \geq 1$, $P(n)$ is true.

Show that $P(1)$ is true:

To establish $P(1)$, we must show that

$$a_1 = 2 \cdot 5^{1-1}. \quad \leftarrow P(1)$$

Now the left-hand side of $P(1)$ is

$$a_1 = 2 \quad \text{by definition of } a_1, a_2, a_3, \dots,$$

and the right-hand side of $P(1)$ is

$$2 \cdot 5^{1-1} = 2 \cdot 5^0 = 2 \cdot 1 = 2.$$

Thus the two sides of $P(1)$ are equal to the same quantity, and hence $P(1)$ is true.

Show that for each integer $k \geq 1$, if $P(k)$ is true then $P(k + 1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 1$. That is:] Let k be any integer with $k \geq 1$, and suppose that

$$a_k = 2 \cdot 5^{k-1}. \quad \begin{array}{l} \leftarrow P(k) \\ \text{inductive hypothesis} \end{array}$$

[We must show that $P(k + 1)$ is true. That is:] We must show that

$$a_{k+1} = 2 \cdot 5^{(k+1)-1},$$

or, equivalently,

$$a_{k+1} = 2 \cdot 5^k. \quad \leftarrow P(k + 1)$$

But the left-hand side of $P(k + 1)$ is

$$\begin{aligned} a_{k+1} &= 5a_{(k+1)-1} && \text{by definition of } a_1, a_2, a_3, \dots \\ &= 5a_k && \text{since } (k+1) - 1 = k \\ &= 5 \cdot (2 \cdot 5^{k-1}) && \text{by inductive hypothesis} \\ &= 2 \cdot (5 \cdot 5^{k-1}) && \text{by regrouping} \\ &= 2 \cdot 5^k && \text{by the laws of exponents,} \end{aligned}$$

which is the right-hand side of $P(k + 1)$ [as was to be shown].

[Since we have proved the basis step and the inductive step, we conclude that the formula holds for all terms of the sequence.] ■

Principle of Strong Mathematical Induction

Let $P(n)$ be a property that is defined for integers n , and let a and b be fixed integers with $a \leq b$. Suppose the following two statements are true:

1. $P(a), P(a + 1), \dots$, and $P(b)$ are all true. (**basis step**)
2. For every integer $k \geq b$, if $P(i)$ is true for each integer i from a through k , then $P(k + 1)$ is true. (**inductive step**)

Then the statement

for every integer $n \geq a$, $P(n)$

is true. (The supposition that $P(i)$ is true for each integer i from a through k is called the **inductive hypothesis**. Another way to state the inductive hypothesis is to say that $P(a), P(a + 1), \dots, P(k)$ are all true.)



Strong Mathematical Induction and the Well-Ordering principle for the Integers

Example 5.4.1 **Divisibility by a Prime**

Prove Theorem 4.4.4: Any integer greater than 1 is divisible by a prime number.

Solution The idea for the inductive step is this: If you are given an integer greater than 1 that is not itself prime, then it is a product of two smaller positive integers, each of which is greater than 1. By inductive hypothesis, you are assuming that each of these smaller integers is divisible by a prime number, and so, by transitivity of divisibility, those prime numbers also divide the integer you started with.

Proof (by strong mathematical induction):

Let the property $P(n)$ be the sentence

n is divisible by a prime number. $\leftarrow P(n)$

Show that $P(2)$ is true:

To establish $P(2)$, we must show that

2 is divisible by a prime number. $\leftarrow P(2)$

But this is true because 2 is divisible by 2 and 2 is a prime number.

Show that for every integer $k \geq 2$, if $P(i)$ is true for each integer from 2 through k , then $P(k+1)$ is also true:

Let k be any integer with $k \geq 2$ and suppose that

i is divisible by a prime number for each integer
 i from 2 through k . $\leftarrow$ inductive hypothesis

We must show that

$k+1$ is divisible by a prime number. $\leftarrow P(k+1)$

Case 1 ($k + 1$ is prime): In this case $k + 1$ is divisible by a prime number, namely, itself.

Case 2 ($k + 1$ is not prime): In this case $k + 1 = ab$ where a and b are integers with $1 < a < k + 1$ and $1 < b < k + 1$. Thus, in particular, $2 \leq a \leq k$, and so by inductive hypothesis, a is divisible by a prime number p . In addition because $k + 1 = ab$, we have that $k + 1$ is divisible by a . Hence, since $k + 1$ is divisible by a and a is divisible by p , by transitivity of divisibility, $k + 1$ is divisible by the prime number p .

Therefore, regardless of whether $k + 1$ is prime or not, it is divisible by a prime number *[as was to be shown]*.

[Since we have proved both the basis and the inductive step of the strong mathematical induction, we conclude that the given statement is true.]

Example 5.4.2 Proving a Property of a Sequence with Strong Induction

Define a sequence $s_0, s_1, s_2, \dots$ as follows:

$$s_0 = 0, \quad s_1 = 4, \quad s_k = 6s_{k-1} - 5s_{k-2} \quad \text{for every integer } k \geq 2.$$

- a. Find the first four terms of this sequence.
- b. It is claimed that for each integer $n \geq 0$, the n th term of the sequence has the same value as that given by the formula $5^n - 1$. In other words, the claim is that all the terms of the sequence satisfy the equation $s_n = 5^n - 1$. Prove that this is true.

Solution

- a. $s_0 = 0, \quad s_1 = 4, \quad s_2 = 6s_1 - 5s_0 = 6 \cdot 4 - 5 \cdot 0 = 24,$
 $s_3 = 6s_2 - 5s_1 = 6 \cdot 24 - 5 \cdot 4 = 144 - 20 = 124$

- b. To use strong mathematical induction to show that every term of the sequence satisfies the equation, the basis step must show that the first two terms satisfy it. This is necessary because, according to the definition of the sequence, computing values of later terms requires knowing the values of the *two* previous terms. So if the basis step only shows that the first term satisfies the equation, it would not be possible to use the inductive step to deduce that the second term satisfies the equation. In the inductive step you suppose that for an arbitrarily chosen integer $k \geq 1$, all the terms of the sequence from s_0 through s_k satisfy the given equation and you then deduce that s_{k+1} must also satisfy the equation.

Proof:

Let $s_0, s_1, s_2, \dots$ be the sequence defined by specifying that $s_0 = 0$, $s_1 = 4$, and $s_k = 6a_{k-1} - 5a_{k-2}$ for every integer $k \geq 2$, and let the property $P(n)$ be the formula

$$s_n = 5^n - 1. \quad \leftarrow P(n)$$

We will use strong mathematical induction to prove that for every integer $n \geq 0$, $P(n)$ is true.

Show that $P(0)$ and $P(1)$ are true:

To establish $P(0)$ and $P(1)$, we must show that

$$s_0 = 5^0 - 1 \text{ and } s_1 = 5^1 - 1. \quad \leftarrow P(0) \text{ and } P(1)$$

But, by definition of $s_0, s_1, s_2, \dots$, we have that $s_0 = 0$ and $s_1 = 4$. Since $5^0 - 1 = 1 - 1 = 0$ and $5^1 - 1 = 5 - 1 = 4$, the values of s_0 and s_1 agree with the values given by the formula.

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Show that for every integer $k \geq 1$, if $P(i)$ is true for each integer i from 0 through k , then $P(k+1)$ is also true:

Let k be any integer with $k \geq 1$ and suppose that

$$s_i = 5^i - 1 \text{ for each integer } i \text{ with } 0 \leq i \leq k. \quad \leftarrow \text{inductive hypothesis}$$

We must show that

$$s_{k+1} = 5^{k+1} - 1. \quad \leftarrow P(k+1)$$

But since $k \geq 1$, we have that $k+1 \geq 2$, and so

$$\begin{aligned} s_{k+1} &= 6s_k - 5s_{k-1} && \text{by definition of } s_0, s_1, s_2, \dots \\ &= 6(5^k - 1) - 5(5^{k-1} - 1) && \text{by definition hypothesis} \\ &= 6 \cdot 5^k - 6 - 5^k + 5 && \text{by multiplying out and applying a law of exponents} \\ &= (6 - 1)5^k - 1 && \text{by factoring out 6 and arithmetic} \\ &= 5 \cdot 5^k - 1 && \text{by arithmetic} \\ &= 5^{k+1} - 1 && \text{by applying a law of exponents,} \end{aligned}$$

[as was to be shown].

[Since we have proved both the basis and the inductive step of the strong mathematical induction, we conclude that the given statement is true.]

Example 5.4.3

A Sequence That Involves the Floor Function

Define a sequence $a_1, a_2, a_3, \dots$ as follows:

$$a_1 = 0,$$

$$a_2 = 2,$$

$$a_k = 3a_{\lfloor k/2 \rfloor} + 2 \quad \text{for each integer } k \geq 3.$$

- Find the first seven terms of the sequence.
- Prove that a_n is even for every integer $n \geq 1$.

Solution

a. $a_1 = 0$

$$a_2 = 2$$

$$a_3 = 3a_{\lfloor 3/2 \rfloor} + 2 = 3a_1 + 2 = 3 \cdot 0 + 2 = 2$$

$$a_4 = 3a_{\lfloor 4/2 \rfloor} + 2 = 3a_2 + 2 = 3 \cdot 2 + 2 = 8$$

$$a_5 = 3a_{\lfloor 5/2 \rfloor} + 2 = 3a_2 + 2 = 3 \cdot 2 + 2 = 8$$

$$a_6 = 3a_{\lfloor 6/2 \rfloor} + 2 = 3a_3 + 2 = 3 \cdot 2 + 2 = 8$$

$$a_7 = 3a_{\lfloor 7/2 \rfloor} + 2 = 3a_3 + 2 = 3 \cdot 2 + 2 = 8$$

- Let the property $P(n)$ be the sentence “ a_n is even.” We use strong mathematical induction to show that the property holds for every integer $n \geq 1$.

Show that $P(1)$ and $P(2)$ are true: The property is true for $n = 1$ and $n = 2$ because $a_1 = 0$ and $a_2 = 2$ and both 0 and 2 are even integers.

Show that for every integer $k \geq 1$, if $P(i)$ is true for each integer i from 1 through k , then $P(k + 1)$ is also true: Let k be any integer with $k \geq 1$ and suppose that

a_i is even for each integer i with $1 \leq i \leq k$. ← inductive hypothesis

[We must show that a_k is even.] By definition of $a_1, a_2, a_3, \dots$

$a_k = 3a_{\lfloor k/2 \rfloor} + 2$ for every integer $k \geq 3$. ← $P(k + 1)$

Now $a_{\lfloor k/2 \rfloor}$ is even by the inductive hypothesis [because $k \geq 1$ and so $1 \leq \lfloor k/2 \rfloor \leq k$]. Thus $3a_{\lfloor k/2 \rfloor}$ is even [because it is a product of an odd and an even integer], and hence $3a_{\lfloor k/2 \rfloor} + 2$ is even [because a sum of two even integers is even]. Consequently, a_k , which equals $3a_{\lfloor k/2 \rfloor} + 2$, is even [as was to be shown].

[Since we have proved the basis step and the inductive step of the strong mathematical induction, we conclude that the given statement is true.] ■

TEST YOURSELF

1. In a proof by strong mathematical induction the basis step may require checking a property $P(n)$ for more _____ value of n .
2. Suppose that in the basis step for a proof by strong mathematical induction the property $P(n)$ was checked for every integer n from a through b . Then in the inductive step one assumes that for
any integer $k \geq b$, the property $P(n)$ is true for all values of i from _____ through _____ and one shows that _____ is true.
3. According to the well-ordering principle for the integers, if a set S of integers contains at least _____ and if there is some integer that is less than or equal to every _____, then _____.

EXERCISE SET 5.4

1. Suppose $a_1, a_2, a_3, \dots$ is a sequence defined as follows:

$$a_1 = 1, a_2 = 3,$$

$$a_k = a_{k-2} + 2a_{k-1} \quad \text{for each integer } k \geq 3.$$
 Prove that a_n is odd for every integer $n \geq 1$.
2. Suppose $b_1, b_2, b_3, \dots$ is a sequence defined as follows:

$$b_1 = 4, b_2 = 12,$$

$$b_k = b_{k-2} + b_{k-1} \quad \text{for each integer } k \geq 3.$$
 Prove that b_n is divisible by 4 for every integer $n \geq 1$.

3. Suppose that $c_0, c_1, c_2, \dots$ is a sequence defined as follows:

$$c_0 = 2, c_1 = 2, c_2 = 6,$$

$$c_k = 3c_{k-3} \text{ for every integer } k \geq 3.$$

Prove that c_n is even for each integer $n \geq 0$.

4. Suppose that $d_1, d_2, d_3, \dots$ is a sequence defined as follows:

$$d_1 = \frac{9}{10}, d_2 = \frac{10}{11},$$

$$d_k = d_{k-1} \cdot d_{k-2} \text{ for every integer } k \geq 3.$$

Prove that $0 < d_n \leq 1$ for each integer $n \geq 1$.

5. Suppose that $e_0, e_1, e_2, \dots$ is a sequence defined as follows:

$$e_0 = 12, e_1 = 29,$$

$$e_k = 5e_{k-1} - 6e_{k-2} \text{ for each integer } k \geq 2.$$

Prove that $e_n = 5 \cdot 3^n + 7 \cdot 2^n$ for every integer $n \geq 0$.

6. Suppose that $f_0, f_1, f_2, \dots$ is a sequence defined as follows:

$$f_0 = 5, f_1 = 16,$$

$$f_k = 7f_{k-1} - 10f_{k-2} \text{ for every integer } k \geq 2.$$

Prove that $f_n = 3 \cdot 2^n + 2 \cdot 5^n$ for each integer $n \geq 0$.

- H 10. The introductory example solved with ordinary mathematical induction in Section 5.3 can also be solved using strong mathematical induction. Let $P(n)$ be “any n ¢ can be obtained using a combination of 3¢ and 5¢ coins.” Use strong mathematical induction to prove that $P(n)$ is true for every integer $n \geq 8$.

11. You begin solving a jigsaw puzzle by finding two pieces that match and fitting them together. Every subsequent step of the solution consists of fitting together two blocks, each of which is made up of one or more pieces that have previously been assembled. Use strong mathematical induction to prove that for every integer $n \geq 1$, the number of steps required to put together all n pieces of a jigsaw puzzle is $n - 1$.

- H 12. The sides of a circular track contain a sequence of n cans of gasoline. For each integer $n \geq 1$, the total amount in the cans is sufficient to enable a certain car to make one complete circuit of the track. In addition, all the gasoline could fit into the car’s gas tank at one time. Use mathematical induction to prove that it is possible to find an initial location for placing the car so that it will be able to traverse the entire track by using the various amounts of gasoline in the cans that it encounters along the way.

- H 13. Use strong mathematical induction to prove the existence part of the unique factorization of integers theorem (Theorem 4.4.5). In other words, prove that every integer greater than 1 is either a prime number or a product of prime numbers.



Application: Correctness of Algorithms

Assertions

Consider an algorithm that is designed to produce a certain final state from a certain initial state. Both the initial and final states can be expressed as predicates involving the input and output variables. Often the predicate describing the initial state is called the **pre-condition for the algorithm**, and the predicate describing the final state is called the **post-condition for the algorithm**.

Example 5.5.1

Algorithm Pre-Conditions and Post-Conditions

Here are pre- and post-conditions for some typical algorithms.

- a. Algorithm to compute a product of nonnegative integers

Pre-condition: The input variables m and n are nonnegative integers.

Post-condition: The output variable p equals mn .

- b. Algorithm to find quotient and remainder of the division of one positive integer by another

Pre-condition: The input variables a and b are positive integers.

Post-condition: The output variables q and r are integers such that $a = bq + r$ and $0 \leq r < b$.

- c. Algorithm to sort a one-dimensional array of real numbers

Pre-condition: The input variable $A[1], A[2], \dots, A[n]$ is a one-dimensional array of real numbers.

Post-condition: The output variable $B[1], B[2], \dots, B[n]$ is a one-dimensional array of real numbers with same elements as $A[1], A[2], \dots, A[n]$ but with the property that $B[i] \leq B[j]$ whenever $i \leq j$. ■

The steps of an algorithm are divided into sections with assertions about the current state of algorithm variables inserted at strategically chosen points:

[*Assertion 1: pre-condition for the algorithm*]
 {Algorithm statements}
 [*Assertion 2*]
 {Algorithm statements}
 ⋮
 [*Assertion $k - 1$*]
 {Algorithm statements}
 [*Assertion k : post-condition for the algorithm*]

Loop Invariants

The method of loop invariants is used to prove correctness of a loop with respect to certain pre- and post-conditions. It is based on the principle of mathematical induction. Suppose that an algorithm contains a **while** loop and that entry to this loop is restricted by a condition G , called the **guard**. Suppose also that assertions describing the current states of algorithm variables have been placed immediately preceding and immediately following the loop. The assertion just preceding the loop is called the **pre-condition for the loop** and the one just following is called the **post-condition for the loop**. The annotated loop has the following appearance:

```
[Pre-condition for the loop]
while ( $G$ )
    [Statements in the body of the loop.
     None contain branching statements
     that lead outside the loop.]
end while
[Post-condition for the loop]
```

Definition

A loop is defined as **correct with respect to its pre- and post-conditions** if, and only if, whenever the algorithm variables satisfy the pre-condition for the loop and the loop terminates after a finite number of steps, the algorithm variables satisfy the post-condition for the loop.

The following theorem, called the *loop invariant theorem*, formalizes these ideas. It was first developed by C. A. R. Hoare in 1969.

Theorem 5.5.1 Loop Invariant Theorem

Let a **while** loop with guard G be given, together with pre- and post-conditions that are predicates in the algorithm variables. Also let a predicate $I(n)$, called the **loop invariant**, be given. If the following four properties are true, then the loop is correct with respect to its pre- and post-conditions.

- I. Basis Property:** The pre-condition for the loop implies that $I(0)$ is true before the first iteration of the loop.
- II. Inductive Property:** For every integer $k \geq 0$, if the guard G and the loop invariant $I(k)$ are both true before an iteration of the loop, then $I(k + 1)$ is true after an iteration of the loop.
- III. Eventual Falsity of Guard:** After a finite number of iterations of the loop, the guard G becomes false.
- IV. Correctness of the Post-Condition:** If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.

Example 5.5.2

Correctness of a Loop to Compute a Product

The following loop is designed to compute the product mx for a nonnegative integer m and a real number x , without using a built-in multiplication operation. Prior to the loop, variables i and $product$ have been introduced and given initial values $i = 0$ and $product = 0$.

*[Pre-condition: m is a nonnegative integer,
 x is a real number, $i = 0$, and $product = 0$.]*

while ($i \neq m$)

1. $product := product + x$

2. $i := i + 1$

end while

[Post-condition: $product = mx$]

Let the loop invariant be

$I(n): i = n \quad \text{and} \quad product = nx$

The guard condition G of the **while** loop is

$G: i \neq m$

Use the loop invariant theorem to prove that the **while** loop is correct with respect to the given pre- and post-conditions.

Solution

I. Basis Property: *[$I(0)$ is true before the first iteration of the loop.]*

$I(0)$ is “ $i = 0$ and $product = 0 \cdot x$,” which is true before the first iteration of the loop because $0 \cdot x = 0$.

II. Inductive Property: *[If $G \wedge I(k)$ is true before a loop iteration (where $k \geq 0$), then $I(k + 1)$ is true after the loop iteration.]*

Suppose k is a nonnegative integer such that $G \wedge I(k)$ is true before an iteration of the loop. Then as execution reaches the top of the loop, $i \neq m$, $product = kx$, and $i = k$. Since $i \neq m$, the guard is passed and statement 1 is executed. Before execution of statement 1,

$$product_{old} = kx.$$

Thus execution of statement 1 has the following effect:

$$product_{new} = product_{old} + x = kx + x = (k + 1)x.$$

Similarly, before statement 2 is executed,

$$i_{\text{old}} = k,$$

so after execution of statement 2,

$$i_{\text{new}} = i_{\text{old}} + 1 = k + 1.$$

Hence after the loop iteration, the statement $I(k + 1)$, namely, $(i = k + 1 \text{ and } product = (k + 1)x)$, is true. This is what we needed to show.

III. Eventual Falsity of Guard: *[After a finite number of iterations of the loop, G becomes false.]*

The guard G is the condition $i \neq m$, and m is a nonnegative integer. By I and II, it is known that

for every integer $n \geq 0$, if the loop is iterated
 n times, then $i = n$ and $product = nx$.

So after m iterations of the loop, $i = m$. Thus G becomes false after m iterations of the loop.

IV. Correctness of the Post-Condition: *[If N is the least number of iterations after which G is false and $I(N)$ is true, then the value of the algorithm variables will be as specified in the post-condition of the loop.]*

According to the post-condition, the value of *product* after execution of the loop should be mx . But if G becomes false after N iterations, $i = m$. And if $I(N)$ is true, $i = N$ and *product* = Nx . Since both conditions (G false and $I(N)$ true) are satisfied, $m = i = N$ and *product* = mx as required. ■

TEST YOURSELF

1. A pre-condition for an algorithm is _____ and a post-condition for an algorithm is _____.
2. A loop is defined as correct with respect to its pre- and post-conditions if, and only if, whenever the algorithm variables satisfy the pre-condition for the loop and the loop terminates after a finite number of steps, then _____.
3. For each iteration of a loop, if a loop invariant is true before iteration of the loop, then _____.
4. Given a **while** loop with guard G and a predicate $I(n)$ if the following four properties are true, then

the loop is correct with respect to its pre- and post-conditions:

- (a) The pre-condition for the loop implies that _____ before the first iteration of the loop.
- (b) For every integer $k \geq 0$, if the guard G and the predicate $I(k)$ are both true before an iteration of the loop, then _____.
- (c) After a finite number of iterations of the loop, _____.
- (d) If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified _____.

EXERCISE SET 5.5

Exercises 1–5 contain a while loop and a predicate. In each case show that if the predicate is true before entry to the loop, then it is also true after exit from the loop.

1. loop: **while** ($m \geq 0$ and $m \leq 100$)
 $m := m + 1$
 $n := n - 1$
 end while

predicate: $m + n = 100$

2. loop: **while** ($m \geq 0$ and $m \leq 100$)
 $m := m + 4$
 $n := n - 2$
 end while

predicate: $m + n$ is odd

3. loop: **while** ($m \geq 0$ and $m \leq 100$)
 $m := 3 \cdot m$
 $n := 5 \cdot n$
 end while

predicate: $m^3 > n^2$

4. loop: **while** ($n \geq 0$ and $n \leq 100$)
 $n := n + 1$
 end while

predicate: $2^n < (n + 2)!$

5. loop: **while** ($n \geq 3$ and $n \leq 100$)
 $n := n + 1$
 end while

predicate: $2n + 1 \leq 2^n$

Exercises 6–9 each contain a while loop annotated with a pre- and a post-condition and also a loop invariant. In each case, use the loop invariant theorem to prove the correctness of the loop with respect to the pre- and post-conditions.

6. [Pre-condition: m is a nonnegative integer, x is a real number, $i = 0$, and $exp = 1$.]

while ($i \neq m$)
 1. $exp := exp \cdot x$
 2. $i := i + 1$
 end while

[Post-condition: $exp = x^m$]

loop invariant: $I(n)$ is “ $exp = x^n$ and $i = n$.”

7. [Pre-condition: $\text{largest} = A[1]$ and $i = 1$]

while ($i \neq m$)

1. $i := i + 1$

2. **if** $A[i] > \text{largest}$ **then** $\text{largest} := A[i]$

end while

[Post-condition: $\text{largest} = \text{maximum value of } A[1], A[2], \dots, A[m]$]

loop invariant: $I(n)$ is “ $\text{largest} = \text{maximum value of } A[1], A[2], \dots, A[n + 1]$ and $i = n + 1$.”

8. [Pre-condition: $\text{sum} = A[1]$ and $i = 1$]

while ($i \neq m$)

1. $i := i + 1$

2. $\text{sum} := \text{sum} + A[i]$

end while

[Post-condition: $\text{sum} = A[1] + A[2] + \dots + A[m]$]

loop invariant: $I(n)$ is “ $i = n + 1$ and $\text{sum} = A[1] + A[2] + \dots + A[n + 1]$.”

9. [Pre-condition: $a = A$ and A is a positive integer.]

while ($a > 0$)

$a := a - 2$

end while

[Post-condition: $a = 0$ if A is even and $a = -1$ if A is odd.]

loop invariant: $I(n)$ is “Both a and A are even integers or both are odd integers and, in either case, $a \geq -1$.”